



DEALER BEST PRACTICE SERIES

Improving the Last Labor to Work
Order Invoicing (LLI) Process

Application Maintenance Site Management **Component Rebuild** Component Life Management Safety MARC Management

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1.0 Introduction

Last Labor to Invoice (LLI) is a measure of the time in days from the completion of a repair to when the work order is closed and the customer is invoiced. Managing and working to reduce this metric is a critical part of maintaining an efficient service shop or CRC. Excessive LLI delays the timing of cash flows, which can negatively impact a company's financial position. In some cases, capital charges are levied against Service Work In Process (SWIP) that has yet to be invoiced. Eliminating these non-value added charges will lead to improved profitability. Finally, by reducing the time a customer has to wait to be invoiced, increased customer satisfaction can be achieved.

Perhaps the most critical element is driving the awareness of the issue across the service department. There are improvements that can be made to the process, but without the appreciation of the issue, delays can continue to occur when attention is diverted to what is deemed a more important use of time. This Best Practice documents the work of several dealers to determine the root causes and potential solutions to excessive LLI.

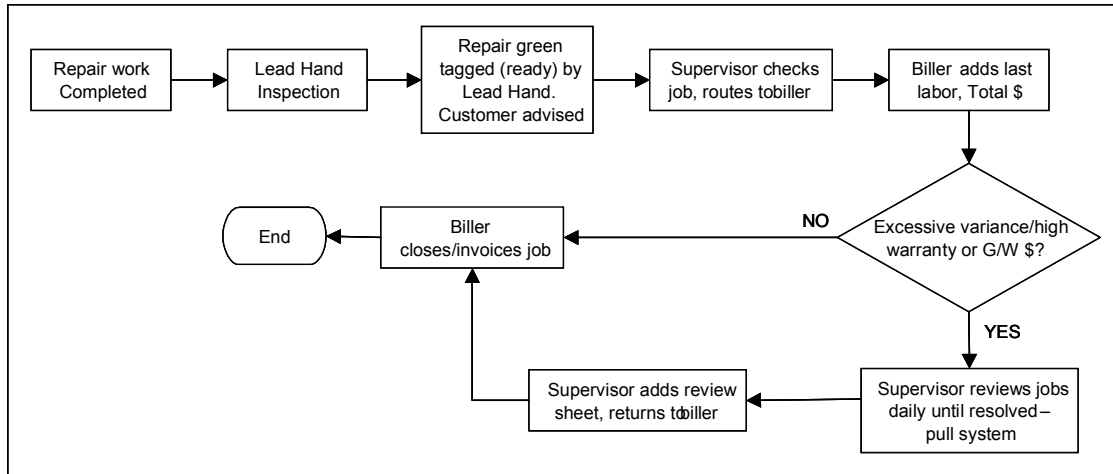
A good benchmarks for LLI is three days. The appropriate goal will vary from dealer to dealer based on factors surrounding obtaining customer purchase order numbers, parts/core return processes, report reviews, and many others.

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2.0 Best Practice Description

It is important to identify the “As-Is” process prior to attempting to make improvements so bottlenecks and other opportunities can be identified and then demonstrated through the “To-Be” process. When all details of the process can be viewed, it is much more likely that problems will be more apparent and give you a baseline for future improvements. The LLI process varies from dealer to dealer, but the following is an example of an improved process implemented by a dealer.



Many LLI processes allow excessive time during the routing and review portions without giving an individual responsibility for ensuring steps are completed in a timely manner.

In the process above, the floor supervisor is responsible for checking over the job upon being tagged as “Ready to Go” by the lead hand and then routing the work order to the biller. Prior to routing to the biller, a Work Order Completion checklist is completed. This can be completed by the technician or a supervisor, but should be done by whoever is desired to have responsibility for ensuring the work order is complete and is received by the biller in a timely fashion. The checklist ensures that all information necessary to close the work order and invoice the customer has been completed.

A sample is shown below.

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Work Order Completion Checklist	
Technician: _____	Work Order No: _____
Last Labor Date: _____	_____
Done?	Task
	1. Perform Final Inspection
	2. Report additional to your lead person or supervisor
	3. Verify Oils/ Lubricants are on Service Reports
	4. Verify all Cores are turned in
	5. Verify all necessary Parts are returned
	6. Verify Service Report is filled out completely and accurately
	A. Make, Model, S/N fields completed?
	B. Any related driven information?
	C. Service Meter reading included?
	D. SIMS reports completed?
	E. Three C's completed? (Complaint, Cause, Correction)
	F. Service Report signed?
	7. Warranty Jobs:
	A. Cat tag filled out & attached to warranty part?
	B. Parts delivered to warranty storage?
	8. Clean up work area
Comments:	
Signature _____ Clerk: _____	
Date: _____ Date: _____	

Upon totaling the work order, if a job is determined to have excessive variance or is a high dollar warranty or goodwill job, the biller routes it to a supervisor with a pull system. This keeps the supervisor focused only on work orders that need attention. Review time can be a major cause of delays in processing work orders when they sit unnoticed, adding unnecessary time to the process. To combat this delay, excessive variance or high dollar warranty and goodwill work orders are required to be reviewed by supervisors on a daily basis to keep the review time to a minimum.

Ensuring that routing and reviewing of work orders is completed in a timely manner is only the beginning of the potential ways to keep LLI to a minimum. Other key behaviors include:

- Ensure accurate labor, parts, and miscellaneous charges in work orders to minimize back outs
- Turn in *complete* service reports the *first* time
- Stay ahead of warranty/goodwill issues
- Utilize DBS Parts Shop Return process to simplify parts return process (See **References**)
- Develop reliable process for obtaining customer P.O. numbers

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- Establish timing criteria to prevent flood of work orders at the end of the month
- Top level management emphasis on importance of timely paperwork
- Work with necessary parties to plan for periods of high workload
- Ensure adequate backup during vacation periods
- Schedule uninterrupted review time for supervisors

There are many potential process improvements, and the deficiency causing the most delay in the process can vary from dealer to dealer or even between dealer branch stores. It is important to analyze the situation from all angles to ensure that all the potential root causes are found. However, perhaps the **most critical step is developing a culture that avoids unnecessary delays in the process**. Simply taking steps to ensure the Service Reports to review do not get filed under a stack of other work or continuing to follow up until the necessary information is gathered will help reduce LLI.

3.0 Implementation Steps

The first step to improving LLI is to create an “As-Is” process map. Once this is complete, identify potential root causes with methods such as a fishbone diagram. An improved process designed to address the identified root causes should then be implemented. An important part of the implementation process is providing training. Without an understanding of the importance of LLI across the organization, it will be difficult to maintain improvements in the long-term. Some general training that is key to reducing LLI includes:

- Service Warranty Policy
- Goodwill Policy
- Parts Warranty Policy
- Service Report Writing
- STW – Time and Daily Work Order Submittal
- Customer Communication and Follow-Up
- Importance of Reducing LLI

The key to a successful implementation is embedding control methods to ensure that any improvements made to the process are followed in the long-term, rather than just falling back into the previous, unsuccessful process. Potential control methods include:

- Weekly DBS reports tracking LLI
- Track work orders by responsible technician
- Daily reports visible to Supervisors
- Develop standard operating procedures

There have been instances in which a dealer showed significant improvement, but without a control method in place, fell back to previous levels of performance. In some cases, the performance was even worse than the original state. No improvement project is complete without a control in place.

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4.0 Benefits

Dealers who implemented similar process improvements to those outlined in this Best Practices have seen results and benefits such as:

- Reduce LLI by upwards of 50%
- Improved timing of cash flows leads to reduced interest expense
- Improved profitability (If capital charges are levied)
- Overall more efficient service operation
- Timely invoicing leads to increased customer satisfaction

5.0 Resources Required

Parts Shop Return Process Detail

A tool that often goes unused in DBS is the Parts Shop Return Process. It helps to simplify the parts return process by allowing the technician to simply choose which parts to return from the list associated with the work order in question. This eliminates additional work looking up the part number or group number again and makes it easier to get the correct number the first time. This also raises a flag to the parts department so they know a returned part is in transit to eliminate unnecessary backorders.

Ensure that this tool is being used – in some cases, it was not even installed in the dealer's DBS environment. See the screenshots below to identify if it is installed. To search for a work order, simply enter the work order number or segment number. If desired, a particular part can also be searched for within a work order.

DBS – SHOP REQ PARTS RETURN MODULE

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Dealer: I2V                      DEALER BUSINESS SYSTEM          DS300041
Store: 00                      Service Work Order              Release ID: 2.3 004

                                Function Name:
(Enter S to select function; CF12 to Return; CF24 to SIGNOFF with joblog)
FUNCTION      DESCRIPTION      FUNCTION      DESCRIPTION
* WORK ORDER SUPPORT *
- MNTWPWOW    Warr W/O Reopen/Inv Sbm  - DSPWOLBR    Employee Labor Inquiry
- MNTWOPP     Product Pgm Maint             - DSPDLRSTR    Dlr Code / Store No Inq.
- MNTWOPPEQ   Product Pgm Equip Maint      - DSPWPWTC     Warranty Type Code Inq
- MNTWOPOLL   Poll Time Clock              - DSPPCCOR     Core Tracking Inquiry
- MNTWOSOS    SOS Feedback Maintenance - DSPPCEMBO    Customer Backorder Inq.
- S MNTPCSRTN Parts Shop Return - DSPWOSLR     Sell Rate Inquiry
                                - DSPWOEMP     Employee Inquiry
                                - DSPWOSOS     SOS Inquiry
                                - DSPPCSRTN    Parts Shop Inquiry
* ON-LINE REPORT REQ. *
- LSTW0INVC   Invoice Print Request      - DSPWOLIN    Time Entry/Recap
- LSTW0BK0    Backout W/O Inv              - MNTWOLPW    Password Maintenance
- LSTWPBK0    Backout Warr Inv
- LSTW0JOB    Job Closing Summary
- LSTW0TMCD   Time Card Print
- LSTPPRPT    Product Pgm Reporting

CF7: Change Break Mode
SHOP REQ PARTS RETURN EXITED

MA  b
12/041

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DBS – SHOP REQ PARTS RETURN MODULE

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None.

8.0 Acknowledgements

This best practice was created by compiling the proposed processes and results of several dealer Six Sigma projects related to reducing Last Labor to Invoice and Service Work in Process.

Projects by the following dealers and Black Belts were referenced:

- Finning Canada
 - Bob Smith
- Hawthorne Pacific
 - Chris Giannaris
- Yancey Brothers
 - John Miller

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